

Git & GitHub - Installation & Setup

Version control, repositories, commits, branches, collaboration and a basic automation workflow.

Goal of this manual

This manual guides you through the complete installation and configuration process, from preparing the environment to final validation. Follow the steps in order and use each verification checkpoint to confirm that the setup was completed correctly.

Before you start

Confirm the supported operating system, runtime/dependencies and version requirements in the current official documentation before installing. The technical checklist below mirrors the validated tool-specific procedure.

Where to run commands

On Windows use PowerShell or Command Prompt; on macOS/Linux use Terminal. After changing PATH or environment variables, close and reopen the terminal before validating again.

Official references

<https://git-scm.com/downloads>

Step-by-step installation and setup

1. Follow this tool-specific step: Acesse git-scm.com e baixe o instalador do seu sistema.
2. Follow this tool-specific step: Windows: execute o instalador. Para uso comum, mantenha a opção que permite Git no terminal/Path.
3. Follow this tool-specific step: Abra NOVO terminal e execute `git --version`.
4. Follow this tool-specific step: Configure nome: `git config --global user.name "Seu Nome"`.
5. Follow this tool-specific step: Configure e-mail: `git config --global user.email "voce@example.com"`.
6. Follow this tool-specific step: Confira: `git config --global --list`.
7. Follow this tool-specific step: Crie git-lab, entre nela e execute `git init`.
8. Follow this tool-specific step: Crie README.md, execute `git add README.md` e `git commit -m "Primeiro commit"`.
9. Follow this tool-specific step: Execute `git log --oneline` para validar o histórico local.

Commands and validation

```
git --version
git config --global user.name "Seu Nome"
git config --global user.email "voce@example.com"
git config --global --list
git init
git add README.md
```

```
git commit -m "Primeiro commit"
```

```
git log --oneline
```

First working use

Complete the first working use described in the steps above and confirm the expected tool-specific result before adding integrations or advanced configuration.

Common problems and fixes

git não reconhecido: reabra o terminal; revise a opção de integração ao PATH.

Commit reclama de identidade: configure user.name e user.email.

Push falha por autenticação: configure o método exigido pelo provedor (token/SSH); isso não é falha da instalação do Git.

Completion criteria

Treat setup as complete when the version/execution checks work and the first use in this manual finishes without errors. Only then move to integrations or advanced configuration.